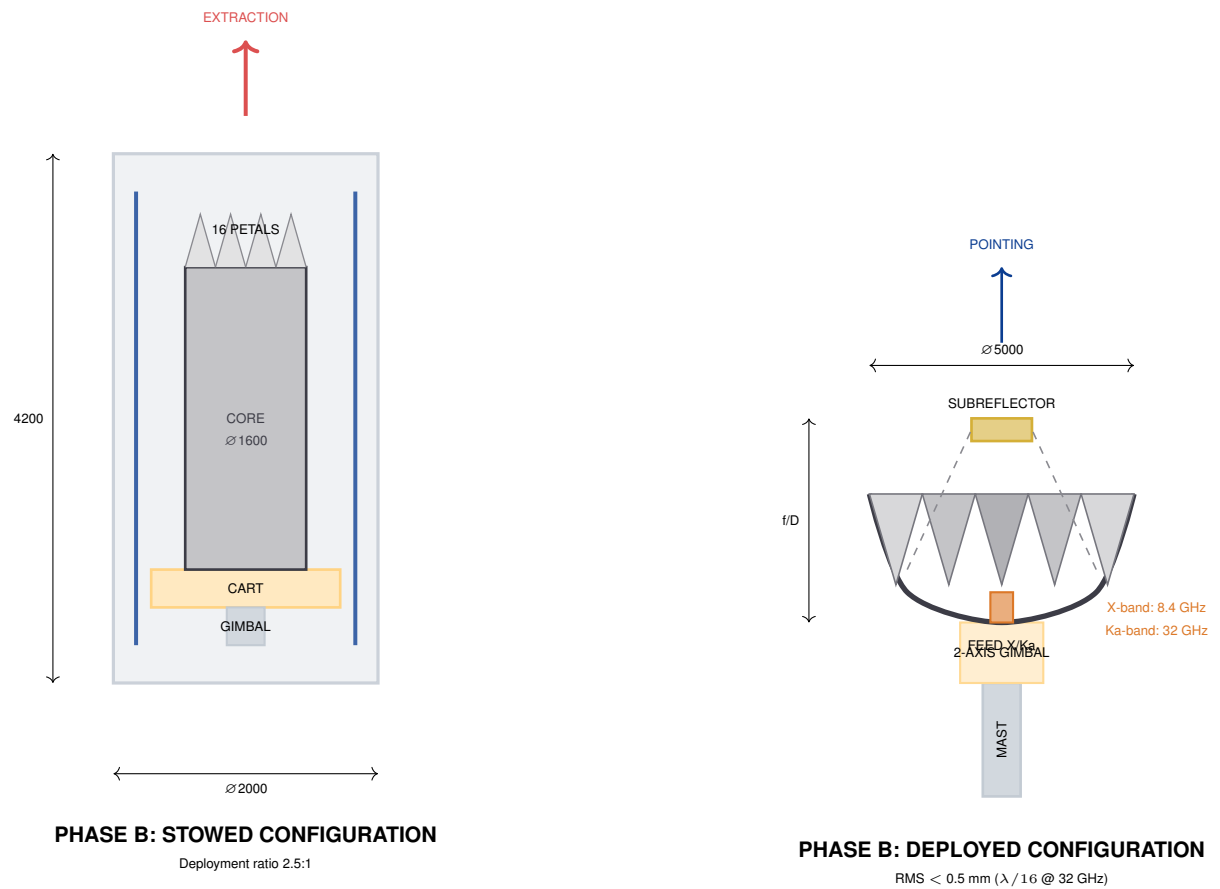


DWG-DSN-01: DSN 5m ANTENNA MODULE — PHASE B

16-Petal CFRP Origami Reflector — Integration into Cylinder DWG-HAB-01



DSN MODULE TECHNICAL SPECIFICATIONS

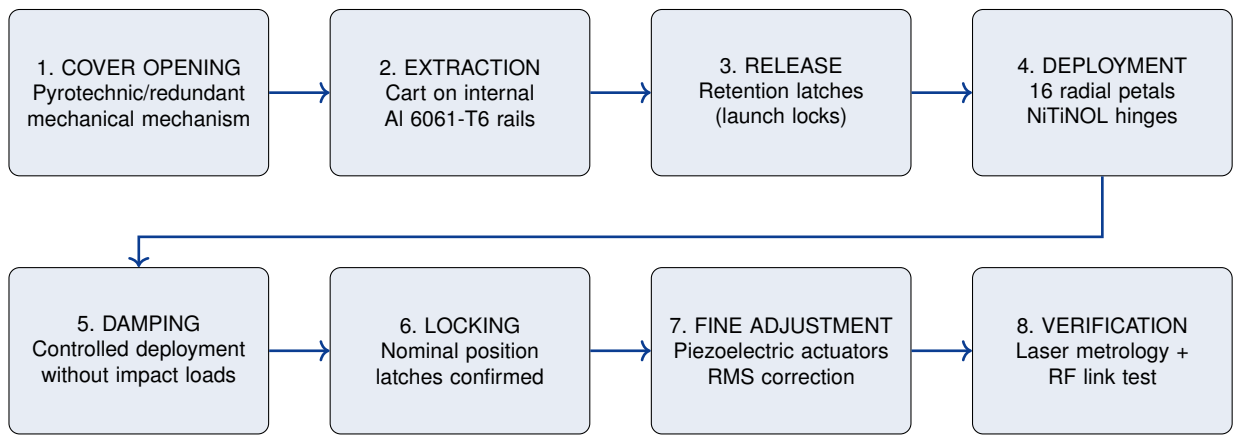
APERTURE AND RF				CRYOGENICS AND ELECTRONICS			
Deployed diameter	5.0 m			HEMT cold stage	4 K (Pulse Tube)		
Stowed diameter	≤ 1.9 m			Preamplifiers	70 K		
Segments	16 CFRP petals			T_n X-band	< 30 K		
Reflector material	CFRP / Cyanate Ester			T_n Ka-band	< 50 K		
Surface precision	< 0.5 mm RMS			Heat rejection	180 W @ 300 K		
Ka-band gain	> 55 dBi			Loop fluid	Propylene glycol/Water		
MECHANICS AND DEPLOYMENT				POINTING AND MASS			
Hinges	NITINOL (tape-spring)			Gimbal	2 axes ($\pm 5^\circ$)		
Fine actuators	Piezoelectric			Stability	< 0.01° (36")		
Launch locks	Redundant, electric			Target mass	205 kg (with margin)		
Extraction rail	Al 6061-T6, 4200 mm			Mass margin	13.8% over 180 kg		
Lubrication	MoS ₂ (sputtering)			Interface	DWG-HAB-01 (internal rails)		

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DRAWING:	DWG-DSN-01 — DSN 5m ANTENNA MODULE (PHASE B)	DATE: 2026-08-11
TITLE:	DSN Origami Antenna 5m — Deployment and Cryogenics	SHEET 1/2
SCALE: 1:25	UNITS: mm	FORMAT: A1 L

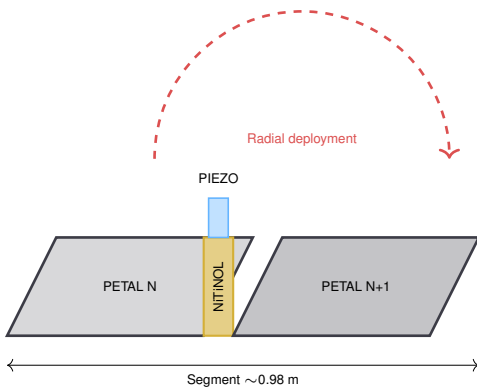
DWG-DSN-01: DEPLOYMENT SEQUENCE AND DETAILS

Origami Kinematics — Thermal Integration with Habitat — EDS-ITO

OPERATIONAL DEPLOYMENT SEQUENCE (PHASE B)



DETAIL A: NITINOL HINGE AND PIEZOELECTRIC ACTUATOR



THERMAL INTEGRATION WITH HABITAT

The heat rejection from the Pulse Tube cryocooler (180 W at 300 K) is transferred to the habitat DWG-HAB-01 via a mechanically pumped fluid loop (MPFL):

- Fluid: Propylene glycol/Water 60/40 (non-toxic, antifreeze).
- Interface: Hermetic feedthrough HAB-007 with cryogenic connectors.
- Habitat dissipation: Plate heat exchanger in ECLSS rack (HAB-009).
- Balance: +180 W (DSN) – 120 W (losses) – 60 W (ECLSS) = positive margin.
- Fail-safe: 50 W backup heaters if cryocooler shuts down.

PSR advantage: The 30–80 K environment reduces the Pulse Tube load, allowing 4 K to be reached with less electrical power than in orbit.

Mitigated thermal risk: Regulating bypass valve prevents habitat overheating if thermal demand falls (e.g., crew absent).

EDS-ITO SYSTEM (LUNAR DUST PROTECTION)

Lunar regolith (JSC-1A) is abrasive and electrostatic. An active Electrodynamic Dust Shields system is integrated on the reflective surface:

- Electrodes: Thin film of Indium Tin Oxide (ITO) transparent to RF.
- Pattern: Interdigitated electrodes at 1–2 mm spacing.
- Voltage: Traveling wave ~500 V_{pp}, 50–100 Hz.
- Target efficiency: >95 % removal without scratching the surface.

Routing on petals: ITO electrodes are deposited on the outer CFRP layer. Connections between adjacent petals use flexible sliding contacts (beryllium copper) that do not compromise RMS precision.

Electromagnetic compatibility: ITO has minimal loss in Ka-band (<0.1 dB). Verification per MIL-STD-461 to avoid interference with feeds.

Reference: NASA TM-2010-216213; Calle et al. 2016.

DETAILED MASS BUDGET (DWG-DSN-01 — PHASE B)

ID	Component	Mass (kg)	Accumulated (kg)
DSN-001	CFRP petals (16 pcs.) + piezo actuators	68.0	68.0
DSN-002	Central core + subreflector + X/Ka feeds	32.0	100.0
DSN-003	NITINOL hinges + redundant latches	18.0	118.0
DSN-004	2-axis gimbal + pointing motors	25.0	143.0
DSN-005	Mast + extraction cart (internal rails)	22.0	165.0
DSN-006	Pulse Tube cryocooler + control electronics	28.0	193.0
DSN-007	100 W X-band TWTA + power supply	12.0	205.0
DSN-008	Thermal loop (pump, piping, exchanger)	8.0	213.0
DSN-009	EDS-ITO system (electrodes + HV supply)	6.0	219.0
DSN-010	RF/fiber wiring + cryogenic connectors	7.0	226.0
Subtotal		226.0	
System margin (13.8 % over 180 kg nominal)		→ 205 kg target	
TOTAL DWG-DSN-01 (with margin)		205.0	

Mass balance note: The Tug L-RAIL budget is 2869 kg. With DWG-HAB-01 (460 kg), Phase A VLBI antenna (515 kg), xenon (8.3 kg), NEXT-C (100 kg) and Tug structure (450 kg), ~1336 kg remain available. Phase B (205 kg) fits comfortably within this remainder, also allowing mass for the VADS and extraction cover.

Next milestones (Phase C/D): Full-scale deployment prototype in cryogenic vacuum (ECSS-E-ST-10-03C); radiation pattern measurement in CATR (IEEE Std 149); qualification of NITINOL hinges per NASA-STD-5017A.

NORMATIVE NOTES AND PHASE B REFERENCES

- ECSS-E-ST-10-03C:** Thermal cycling in vacuum of the deployment mechanism: 3 cycles 293 K ↔ 60 K. Post-cycling functional verification.
- IEEE Std 149-2021:** Radiation pattern and gain measurement in compact range (CATR) at 8.4 GHz and 32 GHz. Acceptance: gain > 55 dBi in Ka.
- NASA-STD-5017A:** Space mechanism design: hinges, latches and actuators with dry MoS₂ lubrication. No metal-metal contact in cryogenic vacuum.
- NASA TM-2010-216213:** Lunar dust mitigation via EDS. Efficiency requirement >95 % without surface degradation.
- Puig, L. et al. (2010):** Review of deployable reflector architectures. Justification of rigid petals vs. membrane/mesh.
- Miura, K. (1985):** Origami folding kinematics applied to 2.5:1 aperture packaging.
- MIL-STD-461:** Electromagnetic interference control for the EDS-ITO system and TWTA/HEMT amplifiers.
- Compatibility with DWG-HAB-01:** Mechanical interface via internal rails (HAB-005), hermetic feedthrough (HAB-007) and shared thermal loop (HAB-010).

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